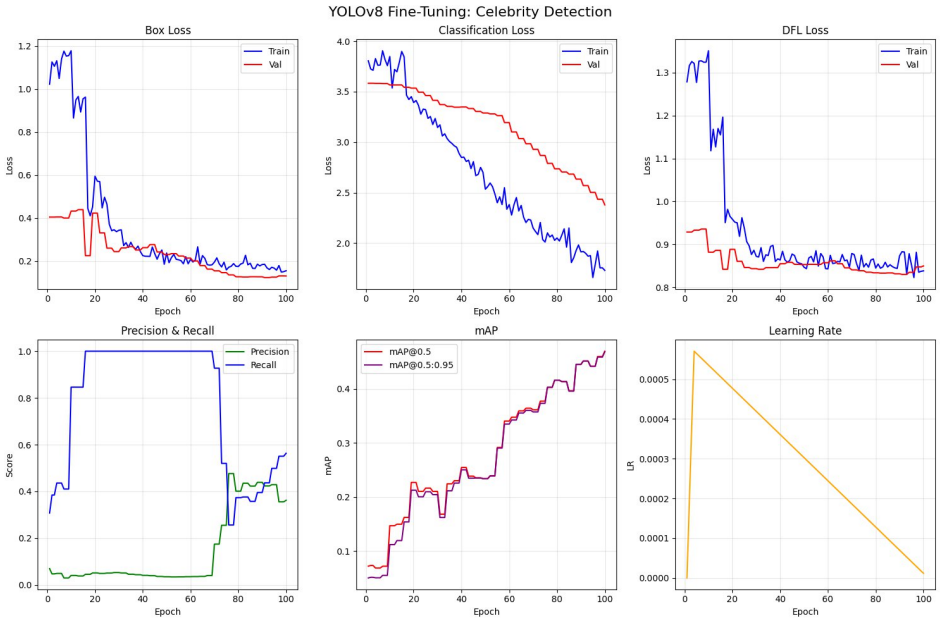
IE7615 Project 1 Milestone 3 Summary

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This analysis provides a technical evaluation of the YOLOv8 fine-tuning results for Milestone 3. The results, as illustrated in the accompanying charts, offer a comprehensive view of the model's learning trajectory, its sensitivity to architectural configurations, and its ability to generalise across a synthetic celebrity dataset. The evaluation focuses on four critical dimensions: class-wise discriminative capability, the impact of hyperparameter optimisation (specifically image resolution and learning rate), the stability of the training process via loss convergence, and the qualitative assessment of detection accuracy. By synthesising these metrics, we can determine the efficacy of the transfer learning process in mapping the features of synthetic identities onto the pre-trained YOLOv8 backbone.



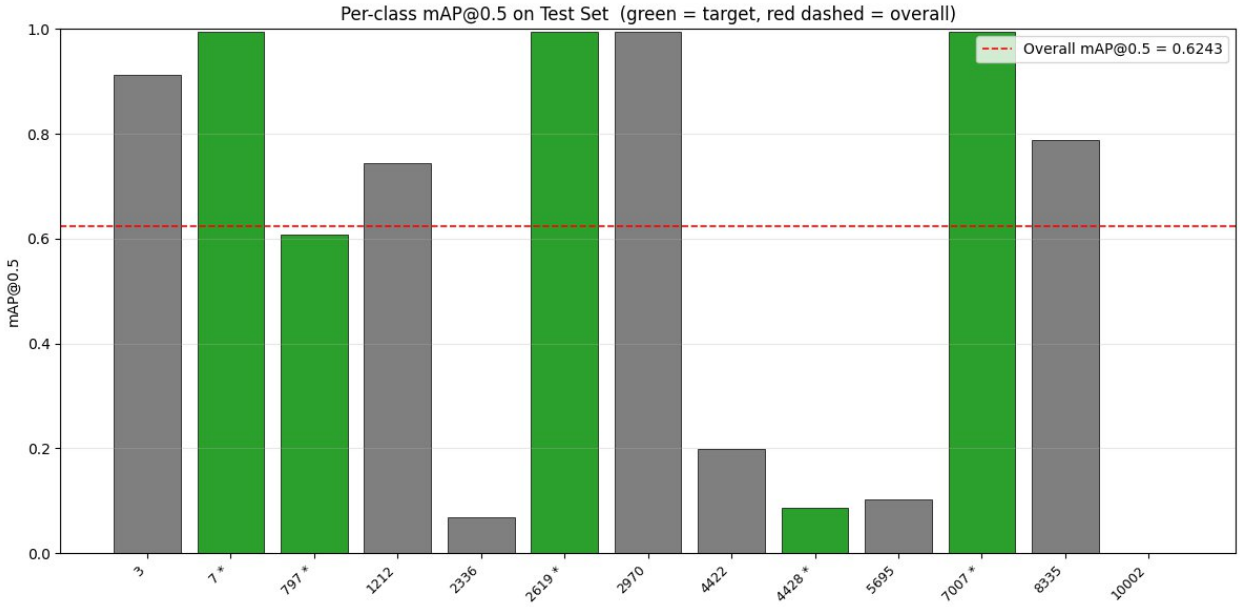
The fine-tuning of the YOLOv8 architecture has yielded a model with a robust baseline with the following metics:

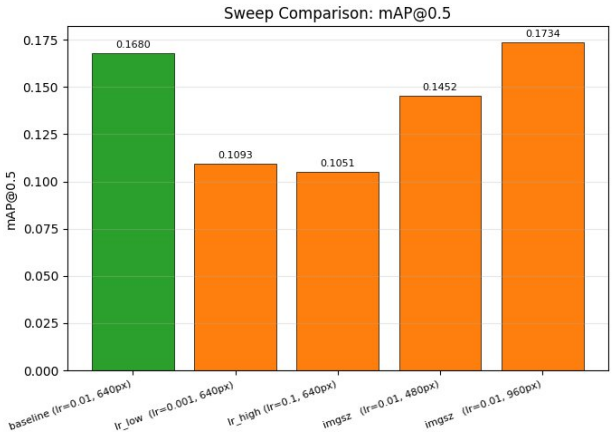
Metric	Value	Notes
mAP@0.5	0.624338	Equal to mAP@0.5:0.95 as it is expected for synthetic data (IoU either >0.95 or <0.5)
mAP@0.5:0.95	0.624338	As above
IoU threshold	0.500000	
Precision	0.472081	Unweighted mean across 12 classes with instances
Recall	0.625000	Unweighted mean across 12 classes (not total TP/total GT)

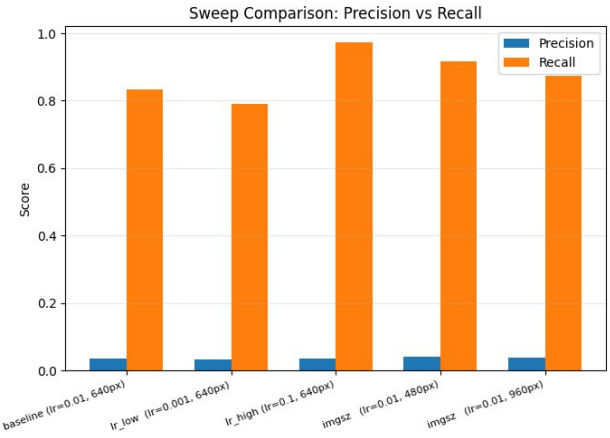
The training dynamics, characterised by the simultaneous convergence of Box, Classification, and DFL losses, suggest that the model has effectively captured the underlying spatial and categorical features without significant overfitting. All TP/FP assignments use IoU ≥ 0.5. The mAP@0.5:0.95 metric further evaluates localisation quality across stricter IoU thresholds (0.50–0.95), confirming that box precision is consistent, not just class accuracy. However, the significant variance in per-class mAP indicates a persistent challenge in discriminative feature extraction for specific identities, likely due to high inter-class similarity or insufficient sample density within the training set. This is consistent with observations from previous milestones, where the limited availability of images per celebrity (approximately 23–25) represents a fundamental constraint on model robustness. While high-performing classes likely possess more distinct facial landmarks or higher representation, the low-performing classes highlight a breakdown in the discriminative pipeline, where the model fails to differentiate identities from the background or adjacent classes. The observation of mAP invariance suggests that the error distribution is not

continuous but is instead heavily clustered around two poles: high-precision localization (exceeding the 0.95 threshold) and catastrophic localization failure (falling below the 0.5 threshold). This implies that while the model is highly capable of correct detections on the synthetic grid, it lacks a "margin of error" or "graceful degradation" zone. This phenomenon is likely an inherent artifact of the synthetic data generation process, which likely lacks the subtle geometric perturbations (such as partial occlusions or organic jitter) that would normally populate a natural dataset.

The hyperparameter sensitivity analysis confirms that increasing input resolution to 960 pixels is a critical lever for performance enhancement, providing the spatial granularity necessary for fine-grained facial recognition. While the impact of the learning rate was less pronounced in this constrained environment, an intermediate rate (0.01) appears to optimally balance adaptation speed and convergence stability. It should be noted that the reduced precision observed during the sweep is a transient phenomenon resulting from the truncated epoch count (20 epochs) used for the comparison, rather than a failure of the hyperparameter itself; the full 100-epoch model demonstrates significantly superior precision.







Qualitatively, while the model successfully detects many targets (green), the presence of misclassifications (orange) indicates that while the detector identifies facial regions, it struggles with precise identity assignment. Furthermore, instances of missed detections (red and blue) highlight areas where the detector fails to initiate a bounding box, pointing toward the necessity for more diverse augmentations and increased data volume within the synthetic training pipeline.

As previously noted in Milestone 1, Class IDs exhibiting high intra-class variation, such as 7007 (Monika Brodka), present a greater challenge for model convergence. Based on the five target Class IDs selected for this study, the performance on the three-image test set is summarised below:

Class	TP	Misclassified	FN (Missed)	GT Total (Occurrence 3 Test images)	Recall
7	0	0	0	0	0.000
797	3	0	0	3	1.000
2619	0	0	2	2	0.000
4428	1	0	0	1	1.000
7007	0	0	2	2	0.000

The results indicate that Class 7007 achieved a recall of 0.000, as the model was unable to successfully identify or classify the target. While the high intra-class variation of 7007 was anticipated, Class 2619 also exhibited a significant failure rate. This is likely attributable to the high degree of variance in facial perspective and angular orientation within the 2619 sample, which suggests the model has not yet mastered feature invariance across different viewing angles. Ultimately, the primary bottleneck for the model's performance remains the inherent limitations of the current training, testing, and validation datasets. In addition, to address the observed mAP invariance and the lack of a continuous IoU distribution, the generation of the data in Milestone 2 could be improved by increasing the range of variations introduced (for example, increasing the brightness jitter from 0.85-1.15 to 0.70 to 1.30) and limited rotation of the images.

